Midterm Study Guide

EE 513 Image Formation and Computational Imaging, Fall 2026

This guide is meant to help you prepare for the midterm, which the syllabus calls Exam 1. For each lecture I've listed the main ideas, some common mistakes, and some practice questions. Near the end there is a section on what the exercises and Project 1 added, and a section on how the lectures connect to one another.

About the Exam

- Covers Lectures 1 through 9, Exercises 1 through 5, and Project 1
- Roughly 30 short-answer questions, one point each
  - Each question asks one thing
  - Most can be answered in two or three sentences
  - Many of them start with "why"
- A few ask for a small calculation, a short derivation, or reading a few lines of code
  - You won't need a calculator. I chose numbers that are easy to work with by hand.
- Many questions ask you to use an idea from one lecture to explain something in another
- I'm more interested in whether you understand why something is true than whether you can recall it
- You may bring one page of hand-written notes
- No calculator, no network access, and no AI

Depth Tiers

Every lecture was labeled with a tier, and the exam only asks of each lecture what its tier promised.

Tier	Lectures	What you should be able to do
Derive	2, 3, 5, 8, 9	Reproduce the derivation, do small calculations, and state the assumptions
Apply	4, 6, 7	Choose a method, explain how it can fail, and say how you'd detect it
Survey	1	Answer short conceptual questions

How to Study

- Write out an answer to each practice question before you look anything up
- If you can't explain why something is true, that's what you need to study
- For each idea, ask yourself where else it came up in the course

1 The Imaging Pipeline (Lecture 1, Survey)

Main Ideas

- Six stages: lens, sensor, ISP, encoding, display, and use
- Every stage throws information away
  - Nothing downstream can recover what an earlier stage destroyed
- Lens: blur (the PSF), distortion, vignetting, and chromatic aberration
- Sensor: sampling, noise, quantization, one color per pixel (the color filter array), and rolling shutter
- ISP: ten or more stages, and the order matters
  - Tuned by people for what they prefer, not for accuracy
- Encoding: JPEG discards what it predicts you won't notice
  - A detector often notices
- A pipeline tuned for people isn't tuned for a machine

Practice

1. Why can't any later processing restore a highlight that clipped at the sensor?
2. Apple ProRAW is called raw. Why won't it work for Project 2?

2 Geometric Image Formation and Calibration (Lecture 2, Derive)

Main Ideas

- Pinhole camera
  - Similar triangles give $x = fX/Z$ and $y = fY/Z$
  - Placing the image plane in front of the pinhole removes the inversion
- Homogeneous coordinates

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

  - Everything is a matrix multiplication except the final division by the third coordinate
  - That division is what destroys depth. Every point along a ray lands on the same pixel.
- Coordinate frames
  - Camera: (X, Y, Z) in meters, with the origin at the optical center, Z pointing into the scene, and Y pointing down
  - Image plane: (x, y)
  - Pixels: (u, v), counted from the top-left corner

- The intrinsic matrix

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \qquad f_x = \frac{f}{p_w}$$

  - f_x is the focal length in pixels, where p_w is the pixel pitch
  - (c_x, c_y) is the principal point
  - The skew s is zero for modern sensors, so four numbers describe the inside of the camera
  - Changing f_x scales the image about the principal point. Changing (c_x, c_y) shifts it.
- Lens distortion

$$x_d = x_n \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) + 2p_1 x_n y_n + p_2 \left(r^2 + 2x_n^2\right)$$

  - Written in normalized coordinates, $x_n = X/Z$ and $y_n = Y/Z$, so the coefficients describe the lens and not the sensor
  - Distortion is applied first, then K
  - The radial terms have only even powers of r and no constant, so distortion is zero at the center and symmetric about it
  - The sign of k_1 determines whether it's barrel or pincushion distortion
  - The tangential terms come from a lens that isn't parallel to the sensor
  - Fit the fewest coefficients the residual justifies
- Correcting distortion resamples the image, which costs some sharpness (Lecture 9)
  - It matters for metrology, stereo, stitching, and wide-angle lenses
  - It rarely matters for classification
- Zhang's method
  - A planar target ($Z = 0$) gives one homography per view, $H = \mu K[r_1\ r_2\ t]$
  - r_1 and r_2 are orthonormal, so each view gives two linear equations in $B = K^{-\top}K^{-1}$:

$$h_1^\top B\, h_2 = 0, \qquad h_1^\top B\, h_1 = h_2^\top B\, h_2$$

  - B is symmetric and only defined up to scale, so it has five unknowns
  - $2n \geq 5$, so you need at least three views
  - OpenCV then refines everything, including distortion, with Levenberg-Marquardt
- RMS reprojection error
  - Measured in pixels
  - It's the quantity the optimizer minimized, so of course it's small
  - It tells you the model fits your data. It doesn't tell you that the data constrained the model.
- Degenerate views
  - Views with the same orientation repeat the same two equations
  - The system stays rank deficient no matter how many views you take
  - In class, nearly parallel views gave a focal length that was 10.7% off, with the same RMS error as a good set of views
  - The standard deviation of the focal length (143 px versus 2.7 px) was the only output that noticed
  - Tilting the board matters more than the number of views

- Signs of a bad calibration
  - Large standard deviations on the parameters
  - f_x and f_y more than a percent apart on a sensor with square pixels
  - A principal point far from the center of the image
  - Coefficients that change a lot when you drop one view
  - Residuals that show a pattern when you plot them as arrows at their image positions

Common Mistakes

- Assuming a small reprojection error means the calibration is right
- Confusing the focal length in millimeters with f_x in pixels
- Fitting more distortion coefficients than the data can support, and then trusting the model in parts of the frame where no corners were detected
- Giving OpenCV the image size as (rows, columns) when it expects (width, height)

Practice

1. Why does Zhang's method require at least three views?
2. Why does the radial distortion model contain only even powers of r?
3. A camera has a 6.0 mm lens and 1.5 μm pixels. What should a calibration report for f_x?
4. Why would you distrust a calibration of a 4000×3000 camera that returned $c_x = 2600$?
5. Why is distortion modeled in normalized coordinates rather than in pixels?
6. Why is a plot of the residuals at their image positions more informative than the RMS reprojection error?

3 Optics, the PSF, and the Frequency Domain (Lecture 3, Derive)

Main Ideas

- The point spread function (PSF) is the image of a point
  - It's the impulse response of the optics
- f-number
  - $N = f/D$, where D is the diameter of the aperture
  - A smaller N is a larger aperture
  - Stopping down means closing the aperture, which increases N
- Diffraction
  - The focal plane contains the squared magnitude of the Fourier transform of the aperture
  - For a circular aperture this is the Airy pattern, and its first dark ring is at $r_{\text{Airy}} = 1.22\,\lambda N$
  - The aperture also sets a hard cutoff. No spatial frequency above $\nu_{\text{c}} = 1/(\lambda N)$ reaches the sensor.
  - The cutoff depends only on the wavelength and the f-number, not on the focal length, the sensor size, or the number of pixels

  - The diffraction MTF is the autocorrelation of the aperture, and it's exactly zero above ν_c
- Contrast is $m = (I_{max} - I_{min})/(I_{max} + I_{min})$
- Linear and shift invariant
  - If the system is linear and the PSF is the same everywhere, imaging is convolution, $g = f * h$, and in the frequency domain $G = F \cdot H$
  - Neither assumption is exactly true. Aberrations grow off axis and sensors saturate. We assume both over a region of the image.
- Transfer functions
  - The optical transfer function is $H = \mathcal{F}\{\text{PSF}\}$
  - The MTF is $|H(u,v)|/H(0,0)$, the fraction of contrast that survives at each frequency
  - A wider PSF has a narrower MTF
  - A Gaussian PSF has a Gaussian MTF that never reaches zero
  - A defocused lens (a uniform disk) has an H that goes negative, which reverses contrast. The MTF hides the sign.
- Reading an MTF curve
  - MTF50 is the most common single number
  - The MTF at Nyquist tells you how much aliasing to expect
  - Always check the units: cycles/mm, line pairs/mm, or cycles/pixel
- The 2-D DFT
  - Low frequencies are at the center after `fftshift`
  - Horizontal structure in the image shows up as vertical structure in the spectrum
  - An edge produces a line through the origin, perpendicular to the edge
  - Periodic patterns produce isolated bright spots
- The phase carries the picture
  - Natural images all have similar magnitude spectra, roughly $1/f$
  - The phase tells you where things are
- Stages in series multiply
  - $\text{MTF}_{\text{system}} = \prod_i \text{MTF}_i$
  - So the system is worse than its worst stage
  - A pixel integrates over its area, a box of width p, so its MTF is $|\operatorname{sinc}(\nu p)|$, which is $2/\pi \approx 0.64$ at Nyquist
  - In class, the lens at 0.54, the pixel at 0.64, and hand shake at 0.84 gave a system MTF of 0.29 at Nyquist
- A phone camera is not diffraction limited
  - At f/1.78 with 2.42 μm pixels, $\nu_c \approx 1021$ cycles/mm, and the Nyquist frequency is 207 cycles/mm
  - Setting $\nu_c = 1/(2p)$ gives the crossover at $N = 2p/\lambda$, about f/8.8
  - So the lens is a weak anti-aliasing filter, and detail above Nyquist comes back as aliasing
  - Three ways to respond: add an optical lowpass filter, let it alias and clean up later, or use smaller pixels
- Aberrations (defocus, coma, and astigmatism) get smaller as you stop down, but diffraction gets larger
  - So most lenses with an adjustable aperture are sharpest two or three stops below their

maximum aperture

Common Mistakes

- Assuming a phone camera is diffraction limited
- Getting "stopping down" backwards. It means a larger N and a smaller aperture. One of the Lecture 3 slides has this backwards.
- Confusing the wavelength of light, λ, with spatial frequency, ν
- Treating contrast above Nyquist as resolution. It's aliasing.
- Forgetting that the MTF is a magnitude, so it can't show contrast reversal

Practice

1. Why doesn't the diffraction cutoff ν_c depend on the focal length?
2. A camera has 2.2 μm pixels and an f/2 lens. For green light ($\lambda = 0.55$ μm), is its resolution limited by diffraction or by sampling?
3. A lens with an MTF of 0.5 at Nyquist is used with a sensor whose pixel MTF is 0.64, and hand shake contributes 0.75. What is the system MTF at Nyquist?
4. Why can't a better sensor fix a soft lens?
5. In class we swapped the phases of two photographs. Why did each result look like the image that donated its phase?
6. As a lens is stopped down from f/1.4 to f/16, why does the image first get sharper and then get softer?

4 Measuring Image Quality (Lecture 4, Apply)

Main Ideas

- A datasheet describes a design. Your camera has its own decentering, hot pixels, and gain.
- Every measurement should answer three questions
  - How good is it?
  - How repeatable is it?
  - How sure are you?
- Pixel count is not resolution
  - The pixels set the sampling grid
  - The lens decides what detail reaches the grid
  - Measuring the MTF in cycles per pixel puts Nyquist at 0.5
- The slanted-edge method (ISO 12233)
  1. Capture a slanted edge in linear light (raw, or undo the gamma)
  2. Find the edge in each row to sub-pixel precision, inside a window
  3. Fit a line through those positions
  4. Project every pixel onto the edge normal and bin at about 1/4 pixel. This is the edge spread function (ESF).

5. Differentiate to get the line spread function (LSF)
6. Window it, take the magnitude of its Fourier transform, and normalize to 1 at DC

- Why the edge is slanted
  - With a tilt of 3 to 7 degrees, each row crosses the edge at a different sub-pixel position
  - Projecting all the rows onto the edge normal gives a supersampled edge profile
- What goes wrong
  - The gamma isn't removed
  - Clipping, which flattens the edge and overstates sharpness
  - An edge angle that's too small or too large
  - Too few rows
  - Measuring a demosaiced edge, which also measures the interpolator. Measure green, or raw.
- A broken estimator still returns a plausible number
  - In class, taking the centroid over the whole row gave 0.053 instead of 0.187 cycles/pixel, with no error
  - Check your code on a synthetic edge with a known answer. A Gaussian PSF of width σ has MTF50 $= \sqrt{\ln 2/(2\pi^2\sigma^2)}$, which is 0.187 cycles/pixel for $\sigma = 1$.
- A number without an uncertainty is not a measurement
  - Repeat the whole measurement
  - Report the mean, the standard deviation, and the method
- SNR versus signal level
  - A slope of 1/2 on a log-log plot where shot noise dominates
  - A slope of 1 where read noise dominates
  - The knee is at $\mu \approx \sigma_r^2$ (Lecture 5)
- Dynamic range is the full well divided by the noise floor, in dB or stops
  - There's more than one definition of the floor, so always state yours
- Vignetting
  - Even a perfect lens falls off as $\cos^4\theta$
  - The ISP corrects it with a gain map, which amplifies the noise in the corners
- Lateral chromatic aberration
  - The lens magnifies the color channels slightly differently
  - Measured as a radial displacement between the channels, and corrected by rescaling them
- JPEG
  - Convert RGB to YCbCr
  - Subsample the chroma, usually 4:2:0, which discards three quarters of it
  - Take the DCT of 8×8 blocks
  - Quantize the coefficients, coarsely at high frequencies
  - Entropy code what's left
  - Chroma subsampling and quantization are the lossy steps
  - The error is structured and concentrated where the detail was
  - Measure from raw. For machine vision, compress once and as late as possible.
- Phase correlation

$$R = \frac{I_1\overline{I_2}}{|I_1\overline{I_2}|}, \qquad \mathcal{F}^{-1}\{R\} = \delta(x - \Delta x,\, y - \Delta y)$$

  - A shift is a linear phase ramp in the frequency domain
  - Window the images first
  - It needs texture with a broad spectrum
  - It only handles translation
  - In class, "sub-pixel" meant about 0.2 pixel

Common Mistakes

- Measuring the MTF from a JPEG or a demosaiced image
- Reporting MTF50 without units, the method, or an uncertainty
- Assuming a plausible result is a correct one

Practice

1. Why must the slanted-edge measurement be made in linear light?
2. Why does a clipped highlight make a lens look sharper than it is?
3. Why is a single straight edge a poor target for phase correlation along the direction of the edge?
4. Two careful engineers report 11.7 and 9.5 stops of dynamic range for the same sensor. How can they both be right?
5. Why can chroma subsampling hurt an inspection system even when the image looks fine?

5 Sensors and Noise (Lecture 5, Derive)

Main Ideas

- Photons become electrons, and electrons become digital numbers
  - The quantum efficiency η is typically 0.5 to 0.9
  - The full well N_{sat} is the most electrons a pixel can hold
  - The conversion gain K is in DN per electron
- Shot noise is physics, not a defect
  - Photon counts are Poisson, so $\text{Var}[N] = \text{E}[N] = \mu$
  - Know how to derive this. $\text{E}[N(N-1)] = \mu^2$, so $\text{E}[N^2] = \mu^2 + \mu$.
- Other sources of noise
  - Read noise σ_r is Gaussian, independent of the signal, and typically 1 to 5 electrons
  - Dark current grows with exposure time and roughly doubles every 6 to 8 °C. It has its own shot noise.
  - Fixed-pattern noise is PRNU (pixel-to-pixel gain) and DSNU (pixel-to-pixel offset). It's the same in every frame.
- The photon transfer curve

$$D = K(N + R) + B \quad \Longrightarrow \quad \text{Var}[D] = K\left(\text{E}[D] - B\right) + K^2\sigma_r^2$$

  - The variance is an affine function of the mean

  - The slope is K, and the intercept c gives $\sigma_r = \sqrt{c}/K$
  - Know how to derive this from the model
- Measuring it
  - Take two flat fields, A and A', at each of many light levels, all at the same ISO
  - The mean is $\frac{1}{2}(\bar{A} + \bar{A}') - B$
  - The variance is $\frac{1}{2}\operatorname{Var}[A - A']$
  - Fit the points below about 70% of saturation, on a small patch at the center
  - Subtract the black level from the mean, but not from the variance
- Why we use the difference of two frames
  - Fixed-pattern noise and lens shading are the same in both frames, so they cancel in the difference
  - The variance of a single frame includes PRNU, which grows as the square of the mean
  - So the points curve upward, and the fitted K (DN per electron) comes out too high
  - The Lecture 5 slide says the bias is low. That's an error. With K in DN per electron the bias is high, as the Exercise 3 solution shows.
- Signal-to-noise ratio

$$\mathrm{SNR} = \frac{\mu}{\sqrt{\mu + \sigma_r^2}} \approx \begin{cases} \sqrt{\mu}, & \mu \gg \sigma_r^2 \text{ (photon limited)} \\ \mu/\sigma_r, & \mu \ll \sigma_r^2 \text{ (read-noise limited)} \end{cases}$$

  - Reducing the read noise only helps below the knee
  - Above the knee, the only way to improve the SNR is to collect more photons
- Consequences
  - Doubling the exposure improves the SNR by $\sqrt{2}$
  - Averaging n frames improves it by $\sqrt{n}$
  - Bigger pixels collect more photons
  - The dynamic range is $N_{\mathrm{sat}}/\sigma_r$
  - Fixed-pattern noise doesn't average away. Remove it by calibration, with dark frames for the offset and flat fields for the gain.
  - Denoising can't create photons
- Rolling shutter
  - The rows are exposed at different times, so moving objects are skewed
  - It breaks the assumption of a single center of projection

Common Mistakes

- Subtracting the black level from the variance
- Including points near saturation in the fit
- Letting the ISO change with the exposure, so the points don't lie on a single line
- Thinking a digital gain, such as white balance or lens shading correction, changes the SNR

Practice

1. Show that a Poisson random variable has a variance equal to its mean.

2. A pixel collects 100 electrons and has a read noise of 10 electrons. Which noise regime is it in?
3. Why does reducing the read noise only improve the SNR below the knee?
4. Why does averaging 16 frames improve the SNR by a factor of 4 rather than 16?
5. What limits the improvement you get from averaging a very large number of frames?
6. After lens shading correction, the corners of a flat field look noisier than the center. Did the correction reduce the SNR in the corners?
7. Why does a sensor add a black-level pedestal rather than reading zero in the dark?

6 Color Science for Cameras (Lecture 6, Apply)

Main Ideas

- A surface reflects a continuous spectrum
  - The eye reduces it to three numbers
  - The camera reduces it to three different numbers
- CIE XYZ
  - Integrates the spectrum against the color matching functions $\bar{x}, \bar{y}, \bar{z}$
  - These are a linear combination of the cone responses, and $\bar{y}$ is luminance
  - It's device independent
  - $L^*a^*b^*$ warps XYZ so that equal distances look roughly equally different
- Metamerism
  - Two different spectra with the same XYZ look identical
  - But only to that observer
  - The camera is a different observer
- The Luther condition
  - A camera is colorimetric only if its spectral sensitivities are a linear combination of the color matching functions
  - No production camera meets it
  - So no matrix can map camera RGB to XYZ exactly for every spectrum
- White balance (von Kries)
  - One gain per channel, so that a neutral patch has equal RGB values
  - Apply it to the mosaic, before demosaicing
  - A large gain amplifies that channel's noise too. Tungsten light needed a gain of 3.4 on blue.
- Color correction matrix
  - A 3×3 least-squares fit from white-balanced camera RGB to XYZ over the 24 patches, so 9 unknowns
  - Rows that sum to 1 preserve neutrals
  - The residual can't be eliminated. It's metamerism showing up as error.
  - The best matrix depends on the spectrum of the light (Exercise 4)
- ΔE
  - Below 1 is imperceptible to most people

  - 1 to 2 is visible on close inspection
  - 2 to 3.5 is visible at a glance
  - Above 5 is obviously a different color
  - Report the mean and the worst patch, and say which metric you used (ΔE_{76} or ΔE_{2000})
- Gamma
  - sRGB is roughly $V = L^{1/2.2}$. It's actually a short linear segment joined to a 1/2.4 power law.
  - A value of 0.5 carries only about 22% of the light
  - Half of the light is encoded as about 0.73
  - Linearize first. Filter, resample, blend, and merge in linear light, and encode at the end.
- Why gamma exists
  - It's perceptual coding, not a correction for CRTs
  - It puts the quantization levels where the eye is most sensitive
  - 8-bit sRGB looks smooth, but 8-bit linear data shows banding in the shadows
  - A 12-bit raw file has enough levels to stay linear

Common Mistakes

- Treating the residual of the color matrix as a bad fit
- Averaging, filtering, or resizing sRGB values directly
- Fitting the white balance gains to a clipped white patch. Use an unclipped mid gray.
- Assuming the CFA pattern is RGGB instead of reading it from the file

Practice

1. Why can't any 3×3 matrix bring every patch to $\Delta E = 0$, no matter how carefully it is fitted?
2. Why is the sRGB average of a fine black-and-white checkerboard about 0.73 rather than 0.5?
3. Why should white balance be applied before demosaicing?
4. Why should you report the worst-case ΔE as well as the mean?

7 The Image Signal Processor (Lecture 7, Apply)

Main Ideas

- The stages, in order: black level, lens shading, defective pixels, white balance, demosaicing, denoising, color matrix, tone curve, sharpening, and encoding
  - The early stages work in linear light and try to recover the scene
  - After the tone curve, the data are display referred and prepared for a viewer
  - Most stages are nonlinear, so they don't commute and the order matters
- PSNR $= 10 \log_{10}(\text{peak}^2/\text{MSE})$
  - 6 dB lower means twice the RMS error
  - 1 dB higher means about 11% less RMS error
- Black level

  - Read noise makes dark pixels swing above and below their dark level, and an ADC can't output negative numbers
  - So the sensor adds an offset, 528 out of 4095 on an iPhone 15 Pro
  - If you leave it in, the shadows are washed out and magenta
  - It cost 5.7 dB in class, compared with 1.2 dB for the best improvement in demosaicing
- Defective pixels
  - Hot or dead pixels
  - Replace each one with the median of its same-color neighbors, so one outlier can't pull the estimate
  - Do this before demosaicing, or one bad pixel becomes a colored blob
- Demosaicing invents two thirds of the image
  - On an RGGB sensor, half the pixels are green and a quarter each are red and blue
  - Bilinear interpolation fills in each channel from its own samples
  - It fails near the sampling limit of the CFA, with zippering and false color
  - These errors are structured and sit on edges
- Malvar, He, and Cutler

$$\hat{G} = \hat{G}_{\text{bil}} + \tfrac{1}{2}\,\Delta_R, \qquad \Delta_R = R - \tfrac{1}{4}\,[\text{the four reds two pixels away}] \approx -\nabla^2 R$$

  - Fine detail is mostly brightness, which all the channels share
  - So the curvature of the measured channel predicts the curvature of the missing one
  - Eight cases, four filter patterns, and one fixed linear filter with no branching
- White balance before demosaicing
  - It was better by 0.6 dB in class, but only after the simulation included cross-talk between the color channels
  - A simulation that leaves out a mechanism can't test a claim that depends on it
- Denoising trades detail for smoothness
  - Adaptive denoisers, such as non-local means, average pixels that look alike rather than pixels that are nearby
- Unsharp masking, $I + k(I - G_\sigma * I)$
  - Applies a gain of $1 + k(1 - \hat{G}_\sigma(f))$ at each frequency
  - It lifts detail that arrived with reduced contrast
  - Where the lens passed nothing, any gain times zero is still zero
  - It causes halos and amplifies noise and aliasing
- Denoise early and sharpen late
  - Real denoisers are nonlinear and depend on the noise model of the sensor
  - Sharpening and the tone curve break that model
- Tuning is about preference, not accuracy
- An ISP for machines
  - Little sharpening, gentle denoising, linear or log output, and fixed white balance
  - Or skip the display stages and give the network calibrated raw data

Common Mistakes

- Forgetting to subtract the black level

- Applying the color matrix after the tone curve
- Sharpening before denoising
- Using a demosaiced image as the ground truth for testing a demosaicing algorithm

Practice

1. Why are defective pixels replaced before demosaicing?
2. Why is a defective pixel replaced with the median of its same-color neighbors rather than the mean?
3. Why must the color matrix come before the tone curve?
4. Why can't sharpening restore detail above the lens cutoff?
5. Why can sharpening make aliasing worse on a phone?
6. In Exercise 5, why is a quarter-resolution image built from 2×2 blocks of the mosaic a valid ground truth for testing demosaicing?

8 Linear Filtering, Gradients, and Edges (Lecture 8, Derive)

Main Ideas

- Convolution and correlation

$$g[m, n] = \sum_{k} \sum_{\ell} h[k, \ell]\, f[m - k,\, n - \ell]$$

  - Convolution flips the kernel and slides it. Correlation slides it without flipping.
  - They're the same for symmetric kernels and differ in sign for derivative kernels
  - OpenCV's `filter2D` and the "convolution" layers in deep learning actually compute correlation
- Smoothing kernels sum to 1, so flat regions are unchanged. Derivative kernels sum to 0, so flat regions become 0.
- Convolution is commutative and associative
  - A cascade of linear filters is one filter
  - Linear filters can be applied in any order
- Non-causal and even
  - Images have no time axis, so kernels are centered and usually even, $h[-\ell] = h[\ell]$
  - An even kernel has a real $H(\omega)$, so it doesn't shift the image
  - ω is in radians per pixel, and $\nu = \omega/2\pi$ is in cycles per pixel
- Small FIR kernels have broad, smooth frequency responses
  - You can't get a sharp cutoff with a few taps
- Borders
  - Zero, replicate, reflect, or wrap (wrap is what the DFT assumes)
  - Every option invents data
  - Replicate or reflect is usually best

- Separability
  - A kernel that is a column times a row (rank 1) can be applied in two passes
  - K^2 multiplications per pixel become $2K$, because the row-filtered image is reused
  - Box, Gaussian, binomial, and Sobel kernels are separable. A disk is not.
- Smoothing kernels
  - A box filter's response goes negative, which reverses contrast
  - A Gaussian's response is another Gaussian, so it's never negative. It's also separable and isotropic.
  - Truncate a Gaussian at $\pm 3\sigma$, about $6\sigma + 1$ taps
  - Binomials, such as [1 2 1]/4 and [1 4 6 4 1]/16, approach a Gaussian by the central limit theorem, with $\sigma^2 = (K-1)/4$, and are exactly zero at 0.5 cycles per pixel
- Smoothing is not lowpass filtering
  - The ideal lowpass kernel is a sinc, which is infinitely long and rings at every edge
  - Smoothing is judged in the image: no ringing, no shift, and no negative weights
- Differences
  - Forward and backward differences are off by half a pixel
  - The central difference, $\frac{1}{2}(f[n+1] - f[n-1])$, is centered
  - As a kernel, it's $\frac{1}{2}[1\ 0\ -1]$ for convolution and $\frac{1}{2}[-1\ 0\ 1]$ for `filter2D`
- A derivative is not a highpass filter
  - Its response is $H = j\omega$, which rises in proportion to frequency, with no flat passband
  - A central difference gives $j \sin \omega$, which falls back to 0 at π
  - Images are strongest at low frequencies and noise is white, so derivatives amplify noise. Smooth first.
- Gaussian derivative
  - $\frac{d}{dx}(G_\sigma * f) = G'_\sigma * f$, so smoothing and differentiating is one kernel
  - Its response, $j\omega\, e^{-\sigma^2\omega^2/2}$, is a bandpass filter with a peak at $\omega = 1/\sigma$
  - σ sets the scale
  - Sobel ([1 2 1] smoothing across a central difference) is a 3×3 approximation. Prewitt uses [1 1 1].
- Gradient
  - $\nabla f = (f_x, f_y)$, with magnitude $\sqrt{f_x^2 + f_y^2}$ and orientation $\operatorname{atan2}(f_y, f_x)$
  - Points toward brighter values, across the edge
  - f_x and f_y are linear filters, but the magnitude and orientation are not
  - With y pointing down, the derivative at angle φ is $\cos\varphi\, f_x + \sin\varphi\, f_y$
- Laplacian
  - $\nabla^2 f = f_{xx} + f_{yy}$, with kernel $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $H \approx -\omega^2$ at low frequencies
  - Linear, unlike the gradient magnitude
  - Gives a zero crossing at a step edge, where the gradient magnitude gives one peak
  - Zero on a constant slope
  - Gives one extremum on a thin line, where the gradient magnitude gives two peaks
  - Has no direction
  - Its noise gain grows as ω^2 rather than ω, so it is always smoothed first
- LoG and DoG

  - $\nabla^2(G_\sigma * f) = (\nabla^2 G_\sigma) * f$
  - Edges are zero crossings. Blobs of radius $\sqrt{2}\,\sigma$ give the strongest response.
  - From the heat equation, $G_{\rho\sigma} - G_\sigma \approx (\rho - 1)\sigma^2 \nabla^2 G_\sigma$
  - So a difference of Gaussians ($\rho \approx 1.6$) approximates the LoG cheaply, often from blurs that have already been computed
- Edges and scale
  - An edge is a sharp change in brightness in one direction. A corner changes in two.
  - Whether a change counts as an edge depends on σ
- Canny
  - Three criteria: detection, localization, and a single response per edge
  - Stage 1, the Gaussian gradient: σ trades detection against localization
  - Stage 2, non-maximum suppression: keep a pixel only if it's larger than both of its neighbors along the gradient
  - Stage 3, hysteresis: keep pixels above T_H, and pixels between T_L and T_H that are 8-connected to one above T_H. Typically T_L is about 0.4 to $0.5\,T_H$.
  - `cv.Canny` doesn't blur first, and its thresholds are absolute gradient values from a 3×3 Sobel

Common Mistakes

- Calling a derivative a highpass filter
- Forgetting that `filter2D` computes correlation, which flips the sign of a derivative
- Truncating a Gaussian well short of $\pm 3\sigma$
- Saying the gradient magnitude gives a $\pm$ pair at a step edge. The Laplacian does. The gradient magnitude gives two peaks only for a thin line.
- Expecting `cv.Canny` to smooth the image for you

Practice

1. Why doesn't an even kernel shift the image?
2. Why is the binomial kernel [1 4 6 4 1]/16 a reasonable approximation of a Gaussian?
3. Derive the frequency response of the central difference $\frac{1}{2}[1\ 0\ -1]$.
4. Why do pyramids and SIFT use a difference of Gaussians rather than a Laplacian of Gaussian?
5. What problem does non-maximum suppression solve?
6. Why does hysteresis use two thresholds rather than one?

9 Sampling, Resampling, and Pyramids (Lecture 9, Derive)

Main Ideas

- Folding
  - $\cos\big(2\pi(1-\nu)n\big) = \cos(2\pi\nu n)$ for every integer n

  - So ν, $1-\nu$, $\nu+1$, and so on all give the same samples
  - Every frequency folds into 0 to 0.5 cycles per pixel, and the spectrum of the samples repeats every 1 cycle per pixel
- The sampling theorem
  - If $f(x)$ has no content at or above 0.5 cycles per pixel, then $f(x) = \sum_n f[n]\ \mathrm{sinc}(x-n)$ exactly
  - Content above 0.5 lands on a lower frequency, and afterward you can't tell it from real content
- Images are not bandlimited
  - A camera's only anti-aliasing filter is its lens and the area of the pixel
  - So most captures already contain aliasing
- Shrinking by a factor of r
  - Multiplies every frequency by r, so anything above $0.5/r$ cycles per input pixel aliases
  - Blur with a lowpass filter, then keep every r-th pixel
  - A Gaussian with $\sigma \approx r/2$ is a reasonable place to start
  - Every practical prefilter trades blur against aliasing
- Libraries differ
  - `INTER_LINEAR` uses the nearest 2×2 pixels regardless of the factor, so it aliases when shrinking
  - `INTER_AREA` averages blocks, which is better but still leaks
  - Pillow widens its kernel with the factor
  - Resize in linear light
- Resampling is interpolation followed by sampling, $f(x) \approx \sum_n f[n]\, h(x-n)$
- Enlarging by 2
  - Insert zeros, which squeezes the spectrum into 0 to 0.25 and adds a mirror image above 0.25
  - A lowpass filter with a cutoff of 0.25 and a gain of 2 per direction removes the image
  - `pyrUp` inserts zeros and then filters with $2 \times [1\ 4\ 6\ 4\ 1]/16$
- Interpolation kernels
  - Nearest neighbor: a box 1 pixel wide
  - Linear: a tent 2 pixels wide (bilinear in 2-D)
  - Cubic: 4 pixels wide (bicubic in 2-D)
  - Lanczos-3: a windowed sinc, 6 pixels wide
- The tent kernel
  - A box convolved with itself, so $H(\nu) = \mathrm{sinc}^2(\nu)$
  - $(2/\pi)^2 \approx 0.41$ at Nyquist, so it blurs
  - Small sidelobes above Nyquist let faint blocky artifacts through
  - Creases at every sample
- Enlarging doesn't add detail
  - Detail above the Nyquist frequency of the small image isn't in the samples
- Why not use a sinc for images?
  - Images aren't bandlimited and are already aliased
  - The kernel is infinitely long and decays only as $1/x$, so it's expensive and every border matters

  - It rings at every edge (the Gibbs phenomenon)
  - Lanczos windows the sinc to make it practical
- A good kernel is even, short, reproduces flat regions ($\sum_n h(x-n) = 1$), is close to ideal, and doesn't ring much
  - No kernel has all five properties
- Interpolating versus approximating kernels
  - Interpolating kernels pass through every sample: $h(0) = 1$ and $h(n) = 0$ at every other integer. Examples are nearest, linear, cubic, Lanczos, and sinc.
  - Approximating kernels pass near the samples. Examples are the Gaussian, the B-spline, and the binomial in `pyrUp`.
  - Approximating kernels are never negative, don't ring or overshoot, and suppress images better, but they blur a little
  - Use them for masks, noisy data, and pyramids
- Pixel centers
  - When shrinking by r, output pixel i is centered at $r(i + \frac{1}{2}) - \frac{1}{2}$
  - Using ri instead shifts the image by $(r-1)/2$ pixels
- Forward versus inverse warping
  - Forward warping sends input pixels to non-integer positions, which leaves holes and pile-ups
  - Inverse warping interpolates the input at $T^{-1}(x')$ for every output pixel, so each output gets exactly one value
  - Inverse warping is usually simpler, and it's what `warpAffine` and `remap` do
- Resample once
  - Each resampling blurs, and the blur accumulates
  - Combine the warps first, then resample once
  - Shrinking by 2 and then enlarging loses everything above 0.25 cycles per pixel
  - Enlarging by 2 and then returning to the original sample positions is exact with an interpolating kernel
- Scale is about smoothing, not pixel count
  - Once an image is smoothed, it can be shrunk without aliasing
- Gaussian pyramid
  - $G_0 = f$ and $G_{\ell+1} = \text{reduce}(G_\ell)$: blur with [1 4 6 4 1]/16, then keep every other row and column
  - It takes $1 + \frac{1}{4} + \frac{1}{16} + \cdots = \frac{4}{3}$ of the storage of the image
  - The binomial has a gain of 0.25 at the new Nyquist frequency of 0.25, so there's some aliasing and more blur
- Laplacian pyramid
  - $L_\ell = G_\ell - \text{expand}(G_{\ell+1})$ and $L_N = G_N$
  - Reconstruction, $G_\ell = L_\ell + \text{expand}(G_{\ell+1})$, is exact for any blur
  - It's overcomplete
  - Each level is a difference of Gaussians, close to a LoG, covering about one octave
- Multiband blending
  - Blend each Laplacian level using the Gaussian pyramid of the mask, $L_\ell = M_\ell L_\ell^A + (1 - M_\ell) L_\ell^B$, and then rebuild

  - Each band gets a transition about as wide as its own wavelength, so there's no seam and no ghosting
- Coarse to fine
  - A search of ± 64 pixels in G_0 is a search of ± 4 pixels in G_4
  - Used in burst alignment, optical flow, template matching, SIFT, and CNNs

Common Mistakes

- Saying the prefilter removes low frequencies. It removes the high frequencies that would alias.
- Saying an ideal lowpass filter can't be used because it's "IIR." Its kernel is an infinitely long sinc. The problems are computation, borders, and ringing.
- Assuming forward warping is simpler than inverse warping
- Saying the CDF of the binomial converges to a Gaussian. The weights (the pmf) do.

Practice

1. A pattern at 0.7 cycles per pixel is sampled once per pixel. At what frequency does it appear?
2. Why does every practical prefilter for shrinking an image by 2 trade blur against aliasing?
3. Why would you use an approximating kernel for pyramids and masks, even though it blurs?
4. Why does lens undistortion (Lecture 2) use inverse warping rather than forward warping?
5. Why is it called a Laplacian pyramid, when no Laplacian kernel is ever applied?
6. Why does multiband blending avoid both a visible seam and ghosting?

10 Lessons from the Exercises and Project 1

The exercises turned the lectures into measurements, and some of the exam questions draw on what you found. You should be able to explain the reasoning behind each of these.

- Use one camera for each series
  - The rear cameras on a phone are separate cameras, and iPhones switch to the ultra wide when you focus close
  - The `lens_id` assertion turns a silent mix of cameras into a loud failure
- The calibration target must be flat, and you should use the square size you measured
  - If the sheet curls, the calibration absorbs the curl into the focal length
- Get the pixel pitch two independent ways (Exercise 2)
  - $1000\, f_{\text{mm}}/f_{\text{px}}$ from the calibration
  - The crop factor from the 35 mm equivalent focal length
  - If the two routes share an input, such as the focal length in the EXIF data, agreement between them can't catch an error in that input
- A green plane of the mosaic is sampled at a spacing of $2p$, so its Nyquist frequency is $1/(4p)$, but the photodiode is still p wide
- Test on a known case first
  - A synthetic Gaussian edge for the MTF
  - A simulated sensor for the photon transfer curve

  - A reference implementation for demosaicing, with agreement to floating-point precision, not just "it looks fine"
- Know what you're varying and what you're holding fixed
  - Calibration views need different tilts and positions
  - Repeated slanted-edge captures should differ only in noise
- The printed target is part of the system
  - If its blur dominates, every camera measures the printer
  - Shooting from twice as far away shrinks the print's blur on the sensor, but not the blur from the lens
- Flat fields
  - Fix the ISO
  - Use exposures longer than about 1/50 s, so screen flicker doesn't differ between the two frames of a pair
  - Use a small patch at the center to avoid lens shading
- Color
  - Check for clipping in the histogram rather than trusting the white level in the file
  - Read the CFA pattern from the file
  - Set the white balance gains from an unclipped mid gray
  - Sample the middle of each patch
  - Three charts give you an uncertainty
  - Falloff that's the same in every round belongs to the lens
- Every number needs an uncertainty
  - That's why `results.json` stores [value, uncertainty] pairs
  - A value that's physically impossible, such as a modulation above 1 at Nyquist or a pixel pitch of a millimeter, means there's a bug

11 How the Lectures Connect

Many of the same ideas came up in several lectures, sometimes under different names. Some of the exam questions ask you to use an idea from one lecture to explain something in another. For each of these, you should be able to name two or more lectures where the idea came up and explain what it predicts.

- Linear shift-invariant systems
  - The lens PSF, the pixel aperture, image filters, interpolation kernels, the Malvar-He-Cutler filter, and unsharp masking are all convolutions
  - The same tools apply to all of them: the impulse response, the frequency response, and the fact that stages in series multiply
- Linear light
  - The slanted-edge method, the photon transfer curve, the color matrix, the early ISP stages, and resizing all assume values that are proportional to light
  - Raw is linear. JPEG is not.
- Aliasing
  - The lens as a weak anti-aliasing filter, the MTF at Nyquist, the lower sampling rate

of red and blue in the mosaic, sharpening, decimation, and the supersampling in the slanted-edge method are all about aliasing

- Gains and noise
  - A gain multiplies the signal and the noise equally, so it changes how visible the noise is, not the SNR
  - The SNR was set by the number of photons and the read noise
  - Filters are different. A derivative or a sharpening kernel changes the balance of signal and noise, because its gain depends on frequency.
- Order of operations
  - Linear filters commute
  - Nonlinear stages, including the tone curve, clipping, and most denoisers, don't
- Scale
  - The width of a Gaussian derivative, the σ in Canny, and the levels of a pyramid all determine what size of structure you respond to
- Plausible but wrong numbers
  - A small reprojection error from degenerate views, a centroid over the whole row in the slanted-edge method, a photon transfer curve from single frames, and a simulation without cross-talk all gave believable answers
  - The defenses are test cases with known answers, independent ways of getting the same number, reported uncertainties, and repeated measurements

Practice

1. Moiré appears on a brick wall in a photo from a phone. What design choices in the camera allowed it?
2. Lens shading correction, white balance, a derivative filter, and unsharp masking all make noise more visible. Which of them reduce the SNR of the image content?
3. Using an example from this term, how could you detect that a method returned a plausible but wrong number?
4. Pick an artifact from this term. How does a filter's frequency response predict it?

Formulas You Should Know

Quantity	**Formula**
Pinhole projection	$x = fX/Z, \quad y = fY/Z$
Focal length in pixels	$f_x = f/p_w$
Calibration views needed	$2n \geq 5 \;\Rightarrow\; n \geq 3$
f-number	$N = f/D$
Airy radius	$r_{\text{Airy}} = 1.22\,\lambda N$

Quantity	**Formula**
Diffraction cutoff	$\nu_{\mathrm{c}} = 1/(\lambda N)$
Sensor Nyquist frequency	$\nu_{\mathrm{Nyq}} = 1/(2p)$; green plane of a mosaic, $1/(4p)$
Diffraction crossover	$N = 2p/\lambda$
Diffraction MTF	$\frac{2}{\pi}\left[\arccos s - s\sqrt{1-s^2}\right], \quad s = \nu/\nu_{\mathrm{c}}$
MTF	$\lvert H(u,v)\rvert / H(0,0)$
Cascade	$\mathrm{MTF}_{\mathrm{system}} = \prod_i \mathrm{MTF}_i$
Pixel aperture MTF	$\lvert\operatorname{sinc}(\nu p)\rvert$, $2/\pi \approx 0.637$ at Nyquist
Gaussian PSF MTF50	$\sqrt{\ln 2/(2\pi^2\sigma^2)}$, 0.187 cycles/pixel for $\sigma = 1$
Contrast	$m = (I_{\max} - I_{\min})/(I_{\max} + I_{\min})$
Phase correlation	$R = I_1\overline{I_2}/\lvert I_1\overline{I_2}\rvert$
Photon transfer curve	$\mathrm{Var}[D] = K(\mathrm{E}[D] - B) + K^2\sigma_r^2$
SNR	$\mu/\sqrt{\mu + \sigma_r^2}$; knee at $\mu = \sigma_r^2$
Dynamic range	$\log_2(N_{\mathrm{sat}}/\sigma_r)$ stops
sRGB (approximate)	$V = L^{1/2.2}$
PSNR	$10\log_{10}(\mathrm{peak}^2/\mathrm{MSE})$
Unsharp masking	$I + k(I - G_\sigma * I)$; gain $1 + k(1 - \hat{G}_\sigma(f))$
Malvar-He-Cutler (green at red)	$\hat{G} = \hat{G}_{\mathrm{bil}} + \frac{1}{2}\Delta_R$
Even kernel response	$H(\omega) = h[0] + 2\sum_{\ell\geq 1} h[\ell]\cos(\omega\ell)$
Derivative; central difference	$j\omega$; $j\sin\omega$
Gaussian derivative peak	$\omega = 1/\sigma$
Laplacian kernel response	$2\cos\omega - 2 \approx -\omega^2$ (per direction)
Binomial variance	$\sigma^2 = (K-1)/4$
Difference of Gaussians	$G_{\rho\sigma} - G_\sigma \approx (\rho - 1)\sigma^2\nabla^2 G_\sigma$
Folding	ν and $1 - \nu$ give the same samples
Shrinking by r	frequency $\times r$; aliases above $0.5/r$
Tent (linear) kernel	$\operatorname{sinc}^2(\nu)$, 0.41 at Nyquist
Laplacian pyramid	$L_\ell = G_\ell - \mathrm{expand}(G_{\ell+1})$